

Log-concavity of codimension-three level Hilbert functions of type two

Abstract

Let $R = k[x_1, x_2, x_3]$, where k is any field. We prove that the Hilbert function

$$h = (1, 3, h_2, \dots, h_e = 2)$$

of every standard graded Artinian level quotient of R of type two is log-concave. The argument uses the minimal resolution of the reindexed graded Matlis dual. Its last syzygy has rank one and full support; this forces every proper subset of the middle columns to be independent over $\text{Frac}(R)$. A putative failure of log-concavity then reduces to a rank-one family of relations, which determines a cyclic Gorenstein submodule. Stanley's characteristic-free theorem for codimension-three Gorenstein Hilbert functions closes the remaining case.

For $j \geq 0$, set

$$N_j = \dim_k R_j = \binom{j+2}{2},$$

and put $N_j = 0$ for $j < 0$. We will use the following characteristic-free form of Stanley's theorem.

Lemma 1 (Stanley). *If B is a standard graded Artinian Gorenstein algebra of embedding dimension at most three and socle degree E , then its Hilbert function is symmetric and is nondecreasing through degree $\lfloor E/2 \rfloor$.*

More precisely, the Hilbert function is an SI-sequence; the monotonicity stated above is the only part needed here. The result holds over an arbitrary field. A characteristic-free proof is given in F. Zanello, *Stanley's theorem on codimension 3 Gorenstein h -vectors*, Proc. Amer. Math. Soc. 134 (2006), 5–8, <https://arxiv.org/abs/math/0411231>.

Theorem 1. *Let $A = R/I$ be a standard graded Artinian level algebra of embedding dimension three, socle degree e , and type two. Then*

$$h_i^2 \geq h_{i-1}h_{i+1} \quad (1 \leq i \leq e-1).$$

The statement holds in every characteristic.

Proof. Put $s = e + 3$. Since A is level of type two with socle degree e , the last free module in its minimal resolution is $R(-s)^2$. Reindex the graded Matlis dual by setting

$$M_d = \text{Hom}_k(A_{e-d}, k), \quad g_d = \dim_k M_d = h_{e-d}.$$

Equivalently, $M \cong \text{Ext}_R^3(A, R(-s))$. Dualizing the minimal resolution of A gives a minimal exact complex

$$0 \longrightarrow R(-s) \xrightarrow{\delta_3} F_2 \xrightarrow{\delta_2} F_1 \xrightarrow{\delta_1} R^2 \longrightarrow M \longrightarrow 0. \quad (1)$$

Thus M is generated by two elements in degree zero. Its socle is one-dimensional and concentrated in degree e . Indeed, for $d < e$, standard generation gives $A_{e-d} = R_1 A_{e-d-1}$, so no nonzero functional in M_d is annihilated by all of R_1 .

Write

$$F_1 = \bigoplus_j R(-j)^{p_j}, \quad F_2 = \bigoplus_j R(-j)^{q_j}.$$

Let $K = \text{Frac}(R)$. Tensoring (1) with K shows that the kernel of δ_2 is one-dimensional. Its spanning vector has no zero coordinate: δ_3 is dual to the first differential in the resolution of A , so its coordinates are precisely the minimal homogeneous generators of I , and each is nonzero. Consequently,

$$\text{every proper subset of the columns of } \delta_2 \text{ is } K\text{-linearly independent.} \quad (2)$$

Fix d with $2 \leq d \leq e$. The corresponding log-concavity inequality is the one centered at g_{d-1} . Define

$$Q_d = \sum_{j \leq d} q_j, \quad P_{<d} = \sum_{j < d} p_j,$$

and let r_d be the K -rank of the restriction of δ_1 to the summands of F_1 having shift strictly less than d . Thus $r_d \in \{0, 1, 2\}$. Finally, set

$$\varepsilon_d = \begin{cases} 1, & Q_d = \text{rank } F_2, \\ 0, & Q_d < \text{rank } F_2. \end{cases}$$

By (2), the columns of δ_2 having shift at most d have rank $Q_d - \varepsilon_d$. Since every differential in a minimal graded free resolution has entries in the homogeneous maximal ideal, these columns can have nonzero entries only in rows of F_1 whose shifts are strictly less than d . Their images lie in the kernel of a rank- r_d map from a $P_{<d}$ -dimensional K -space to K^2 . Therefore

$$Q_d - \varepsilon_d \leq P_{<d} - r_d. \quad (3)$$

Let

$$\Delta^2 g_d = g_d - 2g_{d-1} + g_{d-2}.$$

The Hilbert numerator of (1) is

$$(1-t)^3 \sum_{j \geq 0} g_j t^j = 2 - \sum_j p_j t^j + \sum_j q_j t^j - t^s.$$

Since $d \leq e < s$, summing its coefficients through degree d gives

$$\Delta^2 g_d = 2 + Q_d - P_{<d} - p_d.$$

Together with (3), this yields the central estimate

$$\Delta^2 g_d \leq 2 - r_d + \varepsilon_d - p_d. \quad (4)$$

If $\Delta^2 g_d \leq 0$, then $g_{d-2} + g_d \leq 2g_{d-1}$, and the arithmetic-geometric mean inequality gives $g_{d-2}g_d \leq g_{d-1}^2$. Thus a failure of log-concavity would imply

$$\Delta^2 g_d \geq 1. \quad (5)$$

We now analyze the resulting cases.

First suppose $\varepsilon_d = 1$. Every summand $R(-b)$ of F_2 then has $b \leq d$. By the degree correspondence in the dual resolution, it comes from a minimal generator of I of degree $s - b \geq s - d$. Thus I has no nonzero element in degrees below $s - d$. With $m = e - d + 2 = s - d - 1$, the tested triple is

$$(g_{d-2}, g_{d-1}, g_d) = (N_m, N_{m-1}, N_{m-2}),$$

Since the ratios N_j/N_{j-1} are nonincreasing, this triple is log-concave.

We may therefore assume $\varepsilon_d = 0$. If $r_d = 2$, (4) gives $\Delta^2 g_d \leq 0$, contradicting (5).

If $r_d = 0$, there are no summands of F_1 of shift below d . Indeed, each such summand gives a nonzero minimal relation, so the restricted map would have positive rank otherwise. Equation (3) then gives $Q_d = 0$. Consequently

$$g_{d-2} = 2N_{d-2} = d(d-1), \quad g_{d-1} = 2N_{d-1} = d(d+1),$$

and

$$g_d = 2N_d - p_d = (d+1)(d+2) - p_d.$$

A direct calculation gives

$$g_{d-1}^2 - g_{d-2}g_d = d(2(d+1) + (d-1)p_d) > 0.$$

The only remaining case is $r_d = 1$. From (4) and (5), a failure can occur only if

$$p_d = 0, \quad \Delta^2 g_d = 1. \tag{6}$$

All columns of δ_1 having shift below d span one line over K . Because R is a UFD, this line has a primitive homogeneous generator $v = (v_1, v_2) \in R^2$, whose two components have a common degree $a < d$. The submodule generated by the low-shift columns is then vJ for a homogeneous ideal $J \subseteq R$. To see this, divide one nonzero column by the gcd of its two entries. The resulting primitive vector spans the line, and Euclid's lemma shows that every other column is an R -multiple of it.

Because $p_d = 0$, no relation has shift d , while relations of shift greater than d have no component in degrees at most d . Therefore

$$(\ker(R^2 \rightarrow M))_t = (vJ)_t \quad (t \leq d). \tag{7}$$

Let H be the image of v in M_a and put $n = d - a$.

If $H = 0$, then v itself is a relation of degree $a < d$. Equation (7) and the primitivity of v force $1 \in J$, so $J = R$. In this case set $B_j = 0$ for every j .

Suppose $H \neq 0$. The finite-length module RH has nonzero socle, and this socle is contained in the one-dimensional socle of M . Consequently

$$B = R/\text{Ann}(H)$$

is Artinian Gorenstein, has embedding dimension at most three, and has socle degree

$$E = e - a.$$

For every $j \leq n$ and $c \in R_j$,

$$cH = 0 \iff cv \in \ker(R^2 \rightarrow M)_{a+j} \xrightarrow{(7)} cv \in (vJ)_{a+j} \iff c \in J_j.$$

The last equivalence uses the injectivity of multiplication by the nonzero vector v . Thus J and $\text{Ann}(H)$ agree through degree n .

In both the $H = 0$ and $H \neq 0$ cases, with negative subscripts interpreted as zero, we have for $t = d - 2, d - 1, d$

$$g_t = 2N_t - N_{t-a} + B_{t-a}. \quad (8)$$

Equation (8) gives

$$g_{d-1} \geq N_{d-1}.$$

On the other hand,

$$g_{d-1} = h_{e-d+1} \leq N_{e-d+1}.$$

Since N_j is strictly increasing, it follows that

$$2d \leq e + 2. \quad (9)$$

When $H \neq 0$, this gives

$$2(n - 1) = 2d - 2a - 2 \leq e - 2a \leq e - a = E.$$

When $H \neq 0$, Lemma 1 gives

$$D := B_{n-1} - B_{n-2} \geq 0.$$

For $H = 0$ this also holds, since $D = 0$ by definition.

Set

$$x = g_{d-1} - g_{d-2}.$$

Using (8) and $N_j - N_{j-1} = j + 1$, we obtain

$$x = 2d - n + D = d + a + D \geq d.$$

Also

$$g_{d-2} \leq 2N_{d-2} = d(d - 1).$$

Finally, (6) says that the tested triple has the form

$$(g_{d-2}, g_{d-1}, g_d) = (y, y + x, y + 2x + 1)$$

for $y = g_{d-2}$. Its log-concavity margin is

$$g_{d-1}^2 - g_{d-2}g_d = (y + x)^2 - y(y + 2x + 1) = x^2 - y \geq d^2 - d(d - 1) = d > 0.$$

This contradicts the assumed failure.

We have proved log-concavity of g at every index $d - 1$ with $2 \leq d \leq e$. Reversal preserves the log-concavity inequalities, and $g_d = h_{e-d}$, so the same inequalities hold for h .

The argument is valid over every field. The substantive theorem invoked is the characteristic-free form of Stanley's theorem in Lemma 1; the remaining steps use graded Matlis duality, linear algebra over $\text{Frac}(R)$, and the UFD property of $k[x_1, x_2, x_3]$. $\square$